

THE MOON

38–39 Gravity, motion, and orbits

64–67 Celestial cycles

102–103 The family of the Sun

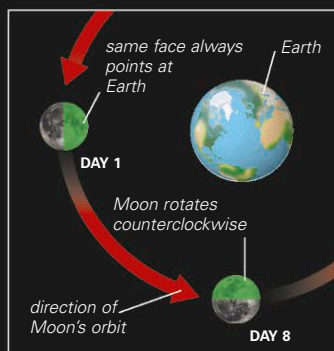
Meteorite impacts 221

EVEN THOUGH IT HAS ONLY 1.2 percent of the mass of Earth, the Moon is still the fifth-largest planetary satellite in the solar system. When full, it is the brightest object in our sky after the Sun, and its gravity exerts a strong influence over our planet.

However, the Moon is too small to retain a substantial atmosphere, and geological activity has long since ceased, so it is a lifeless, dusty, and dead world. Twelve men have walked on its surface and over 838 lb (380 kg) of lunar rock have been collected, but scientists are still not sure exactly how the Moon formed.

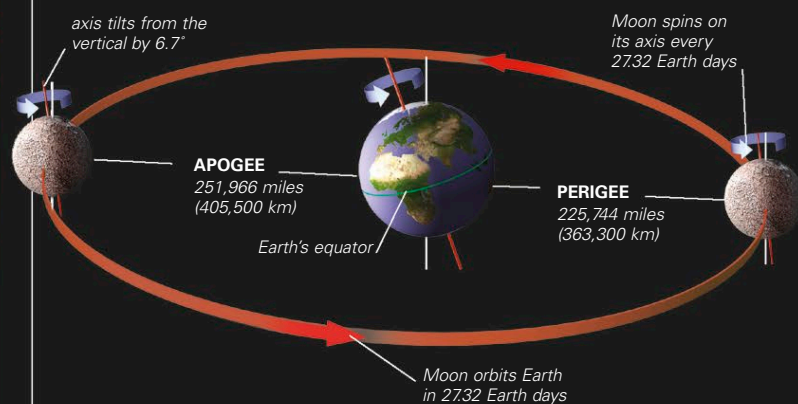
ORBIT

The Moon has an elliptical orbit around the Earth, so the distance between the two bodies varies. At its closest to Earth (perigee), the Moon is 10 percent closer than when at its farthest point (apogee). The Moon takes 27.32 Earth days to spin on its axis, which is the same time it takes to orbit the Earth. This is known as synchronous rotation (see right) and keeps one side of the Moon permanently facing Earth—although eccentricities in the Moon's orbit called librations allow a few regions of the far side to come into view. Because the Earth is moving around the Sun, the Moon takes 29.53 Earth days to return to the same position relative to the Sun in Earth's sky, completing its cycle of phases (see p.66). This is also the length of a lunar day (the time between successive sunrises on the Moon).



SYNCHRONOUS ROTATION

For each orbit of Earth, the Moon spins once on its axis. As a result, it always keeps the same face toward Earth.

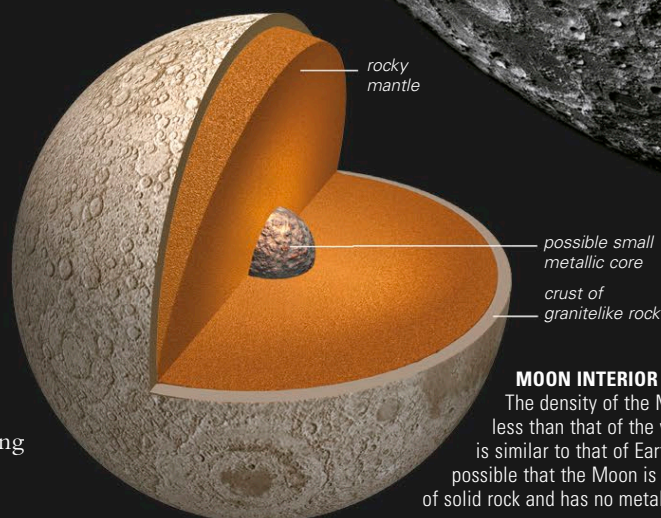


SPIN AND ORBIT

The Moon's orbital path is tilted at an angle to Earth's equator, causing its path across the sky to vary in an 18-year cycle. Tidal forces mean that the Moon is slowing down Earth's rotation, while the Moon moves away from the Earth at a rate of about 1 in (3 cm) each year.

STRUCTURE

The lunar crust is made of calcium-rich, granitelike rock. It is about 30 miles (48 km) thick on the near side and 46 miles (74 km) thick on the far side. Because of the Moon's history of meteorite bombardment, the crust is severely cracked. The cracks extend to a depth of 15 miles (25 km); below that, the crust is completely solid. The Moon's rocky mantle is rich in silicate minerals but poor in metals such as iron. The upper mantle is solid, rigid, and stable. Radioactive decay of minor components of the lunar rock means that the temperature increases with depth. The lower mantle lies about 600 miles (1,000 km) below the crust, and here the rock gradually becomes partially molten. The average density of the Moon indicates that it might have a small iron core. The Apollo missions measured the velocities of shock waves traveling through the Moon, but the results proved inconclusive. Further seismic evidence is needed to confirm the existence of a metallic core.



MOON INTERIOR

The density of the Moon is much less than that of the whole Earth but is similar to that of Earth's mantle. It is possible that the Moon is entirely made of solid rock and has no metallic core at all.